Materials and their uses

The ways a material can be used depend on its properties.

- A saucepan is usually made out of metal and plastic. Metal is used because it is a good conductor of heat. Plastic is used for the handle because plastic is an **insulator**, or poor conductor of heat. This stops you burning your hand when you pick up the saucepan.
- Electrical cables are often made out of a metal called copper because it is a good conductor of electricity and can easily be formed into wires. The copper wire is surrounded with a plastic covering so the electrical cable can be handled safely.
- Ceramic materials are used for lots of different things. Traditional ceramics are used to make dishes and tiles because they are hard, do not absorb liquid and are both heat and rust resistant. Space shuttles have tiles made from ceramics because they protect the shuttle from extreme temperatures.

Scientists are always trying to design better or smarter materials to develop new technology and improve the way objects work. Smart materials are able to respond to changes in the environment.

☞ **1** Complete the table below.

Object	Materials used	Reason why these materials are used
dinner plate		
saucepan		
space shuttle		
electrical cable		

2 Name one piece of equipment you use that your parent/s or guardian/s never had as a child. Describe what it is made from and what it is used for.

Object: _____

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